



Substantial changes in podzol morphology after tree-roots modify soil porosity and hydrology in a tropical coastal rainforest

Pedro Martinez · Peter Buurman · Diego Luciano do Nascimento · Vance Almquist · Pablo Vidal-Torrado

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Abstract

Aims We investigated morphological variations in podzols caused by changes in soil porosity and permeability upon the growth of large tree-roots in a tropical barrier island (Ilha Comprida, Brazil).

Methods Soil morphology was described in a continuous lateral sequence of podzols on a 35 m-long cliff. A soil thin section was used to characterize organic matter (OM) accumulation and to estimate soil porosity, permeability, and saturated hydraulic conductivity (Ksat). Soil texture and the contents of organic carbon, Al, and Fe were determined for each pedogenic horizon containing large tree-roots. The evolution of podzol morphology was interpreted in the context of age determinations by optically stimulated luminescence and ^{14}C .

Results Taproots of cashew trees (*Anacardium occidentale*) penetrated the cemented Bhm horizon and the massive-clayey 2Cgj horizon. Aligned with the taproot, we found a vertical OM-band with lower porosity, permeability, and Ksat than the adjacent Bh and E

horizons. Irregular or broken boundaries between the E and Bh horizons were caused by large tree-roots. While the maximum age of these podzols is 3390 ± 530 years, significant and rapid changes in the Bh-horizon morphology occurred within the lifetime of the cashew trees (~50 years).

Conclusions The interplay between reduction in flow adjacent to large taproots and the enhanced vertical infiltration at depth has resulted into the development of irregular and broken boundaries between the E and Bh horizons. Because tree-roots alter both local soil porosity and water flow paths, they simultaneously cause the formation and degradation of podzol Bh-horizon.

Keywords Podzolisation · Bioturbation · Drainage · Ilha Comprida · *Anacardium occidentale*

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P. Martinez · P. Vidal-Torrado
Departamento de Ciência do Solo, Escola Superior de Agricultura
'Luiz de Queiroz'- ESALQ- USP, Av. Pádua Dias 11, Piracicaba,
SP, Brazil

P. Martinez (✉) · V. Almquist
Department of Crop and Soil Science, Oregon State University,
2750 Campus Way, Corvallis, OR 97331, USA
e-mail: pedro.martinez@oregonstate.edu

P. Buurman
Water Systems and Global Change Group, Wageningen
University, P.O. Box 47, 6700AA Wageningen, The Netherlands

D. L. do Nascimento
ICHNOS Research Group, Departamento de Geologia e Recursos
Naturais, Instituto de Geociências, UNICAMP, Rua João Pandiá
Calógeras, 51, Campinas, SP, Brazil

Introduction

Tree-roots modify soil morphology because they create preferential paths of water flow that affect soil moisture, redox potential, weathering, and translocation of dissolved organic matter (DOM) and mineral grains at different scales (Yatsu 1988; Phillips and FitzPatrick et al. 1999; Phillips and Marion 2004; Gabet et al. 2003; Bengough 2012; Pawlik and Šamonil 2018). As a result, tree-roots contribute to the formation of morphological features (e.g., biopores, textural contrast, and horizon boundary) that persist in the soil for hundreds or even thousands of years (Phillips 2007; Verboom and Pate 2006; Nascimento et al. 2019). The modification of soil morphology driven by tree-roots is easily observed in podzols because podzols have distinct color contrasts between A, E, and B horizons (Schaetzl 1990; Šamonil et al. 2015, 2017, 2018).

The major color contrast in podzols is between a grey-white E horizon depleted in organic matter (OM) and an underlying brown or black B horizon with a high accumulation of OM and metals (Al and/or Fe). Such color contrast is caused by the podzolisation process, which consists in the *vertical* and *lateral* transport of DOM and metals from the surface horizons to the B horizon (DeConinck 1980; Lundström et al. 2000; Buurman and Jongmans 2005; Sauer et al. 2007). The vertical transport of water results in the development of an eluviated E-horizon (DeConinck 1980; Sauer et al. 2007). The accumulation of transported material in the B-horizon is caused by (i) precipitation of organic complexes upon saturation with metals, (ii) decomposition of the organic carrier, and (iii) stagnating water flow (Lundström et al. 2000; Sommer et al. 2001; Buurman and Jongmans 2005; Sauer et al. 2007). DOM can also precipitate because of stagnant groundwater and lateral flow, even in sandy sediment layers with a large porosity, which may result in cemented B horizons and OM-bands (Andriesse 1969, 1970; Buurman et al. 2013; Lopes-Mazzetto et al. 2018). The porosity contrast between a sandy deposit overlying a clayey layer contributes to the accumulation of DOM at the contact (Martinez et al. 2018).

The morphology of podzols is largely controlled by their drainage condition (DeConinck 1980; Mokma et al. 1990, 1994; Buurman et al. 2013). Poorly drained podzols have a flat boundary between the E and B horizons due to the constant presence of the groundwater in the B horizon. The vertical and lateral movement

of groundwater bearing DOM may cause thick and homogeneous dark-brownish Bh horizons in hydromorphic podzols (Martinez et al. 2018). A lush tropical rainforest (e.g., in southeast Brazil) adds high quantities of DOM to the groundwater that moves laterally and results in the formation of B horizons that are locally more than 4 meters thick (Coelho et al. 2010; Buurman et al. 2013; Martinez et al. 2018). Under good drainage, podzols have predominantly vertical water flow and transport of solutes. The downward transport of water results in the deepening of the boundary between the E and B horizon. However, if the water moves through preferential water channels formed by large roots, this leads to irregular boundaries with tongue-like shapes related to the interfingering of the E and B horizons (DeConinck 1980; Buurman and Jongmans 2005; Buurman et al. 2013). In well-drained podzols, OM accumulates as polymorphic OM in pores in the B horizon due to local decomposition of root remains (DeConinck 1980; Buurman and Jongmans 2005; Lopes-Mazzetto et al. 2018), while in poorly-drained podzols monomorphic coatings on sand grains of precipitated OM-Al complexes dominate.

Because soil drainage and hydrological processes are central determinants of podzol morphology, it is expected that root-mediated changes in soil permeability affect the morphological expression of podzolisation. Much of the work that has been done to date has focused on the role of trees in concentrating the flow of vertically infiltrating water. Two prominent examples include the role of pit and mound topography created by tree fall (Šamonil et al. 2018), and the redistribution of precipitation through stemflow and throughfall (Bloomfield 1953; Cornu et al. 1998; Nikodem et al. 2010). In the case of pit and mound microtopography, the pits created by falling trees tend to concentrate precipitation toward the bottom of the depression, resulting in an increase of both E horizon depth and its distinctness (Šamonil et al. 2018). Similarly, stem flow greatly increases the volume of infiltrating water in zones adjacent to the tree. This increase in infiltration results in a vertical expansion of the E and enhanced development of the B horizons, as in the case of the so-called “egg-cup podzols” formed under individual kauri trees (*Agathis australis*) in New Zealand (Bloomfield 1953).

However, for trees to affect the morphology of poorly drained podzols, mechanisms must exist which allow large roots to alter groundwater flow. Trees alter groundwater flow principally through two mechanisms:

(1) transpiration withdrawal of groundwater altering the water table height and flow rates; and (2) the growth and death of (large) roots alter local porosity (Brantley et al. 2017; Sonnenborg et al. 2017). Initially, root growth compresses the soil surrounding the root, resulting in zones of reduced porosity which create vertically oriented capillary barriers (Pawlik et al. 2016b; Brantley et al. 2017; Ruiz et al. 2017). Root death, on the other hand, leaves behind large (often vertical) biopores, which enhance water flow, and consequently the translocation of elements and OM (Burns et al. 1998).

So, although trees have the potential to alter the hydrology of poorly drained podzols, it remains unclear whether there are lasting morphological effects of these localized processes. In part, this is because podzol morphology typically accumulates over decades to millennia (Sauer et al. 2007; Šamonil et al. 2014), which implies that the root effects need to persist over a sufficiently long timescale for a lasting morphological alteration. Based on this premise, large structural roots (several centimeters to a decimeter or larger) are the most likely candidates to strongly affect the macromorphology of poorly drained podzols. This is because these roots often penetrate meters deep and their decay can create zones of preferential flow that remain active for decades to centuries.

Studies on the effects of large structural tree-roots on podzol morphology focused on uprooting (tree-throw) in temperate forests (Schaetzl 1990; Šamonil et al. 2015, 2017, 2018). Tree uprooting can change the shape and sequence of the E and B horizons in podzols while also creating microtopography, and in some cases it alters the pit-scale hydrologic processes (Schaetzl 1990; Phillips and FitzPatrick 1999; Phillips and Marion 2005; Pawlik et al. 2016a, b). Apart from the impacts of uprooting on the morphology and stratigraphy of soil horizons, it is well known that soil drainage conditions determine podzol morphology (DeConinck 1980; Mokma et al. 1990, 1994; Buurman et al. 2013; Jankowski 2014; Lopes-Mazzetto et al. 2018; Martinez et al. 2018). It is still unknown in detail, however, how tree-roots modify local drainage conditions and podzol morphology, especially in tropical regions. Here, we address this issue by asking to what extent tree-root growth changes local drainage and affects the morphology of podzol horizons in a tropical rainforest. We hypothesize that areas in the Bh horizons affected by tree-roots will have larger porosity and greater permeability than the surrounding area. A basic question underlying this work is whether

tree-roots and their mechanical effects on soil physical properties do cause conspicuous vertical and lateral variability of podzol morphology.

To address our central question, we investigated a lateral sequence of initially poorly drained coastal podzols under the tropical rainforest in southeastern Brazil (Fig. 1). The studied sequence of podzols is found in a cliff exposed by rapid erosion in the last half-century. The sequence has marked modifications in Bh-horizon morphology which appear to be triggered by large roots of planted cashew trees (*Anacardium occidentale*). The sequence was studied using a multiscale approach, i.e. from the catena scale (35 m length) to the pore-scale arrangement of OM, voids, and grains (5–200 μm). We described the macromorphology of the lateral sequence, and subsequently the micromorphology of a soil thin section from an area affected by tree-roots, where we analyzed the size and distribution of pores and sand grains, as well as the accumulation of OM. The cross-sections of the pores were measured using binary images generated with ImageJ software. Then, pore dimensions were used to estimate soil porosity, permeability, saturated hydraulic conductivity (Ksat), and capillary pressure. Optically stimulated luminescence (OSL) and ^{14}C age determinations were carried out to provide a chronological framework in which sediment deposition and soil formation can be interpreted.

Materials and Methods

Study area

The studied site is a cliff facing the continent in the southwestern part of the barrier island Ilha Comprida in southeastern Brazil at a latitude of 25°S (Figs. 1 and 2). The Cananeia Sea, a meandering elongated water body connected to estuarine systems separates Ilha Comprida from the continent and delimits it to the SW and NE. Ilha Comprida is a dominantly prograding barrier that has evolved over the last 6000 years by sediment trapping downstream of remnants of an alkaline rock intrusion named Morrete (Fig. 1) (Giannini et al. 2009; Guedes et al. 2011). The studied cliff has recently undergone rapid erosion (100 meters from 1940 to 1970) because it is positioned at the junction of fluvial-marine currents with the ebb tides from the inlet and the Cananeia Sea (Tessler et al. 1990). Accelerated by storm tides, the erosion of this cliff improved the drainage

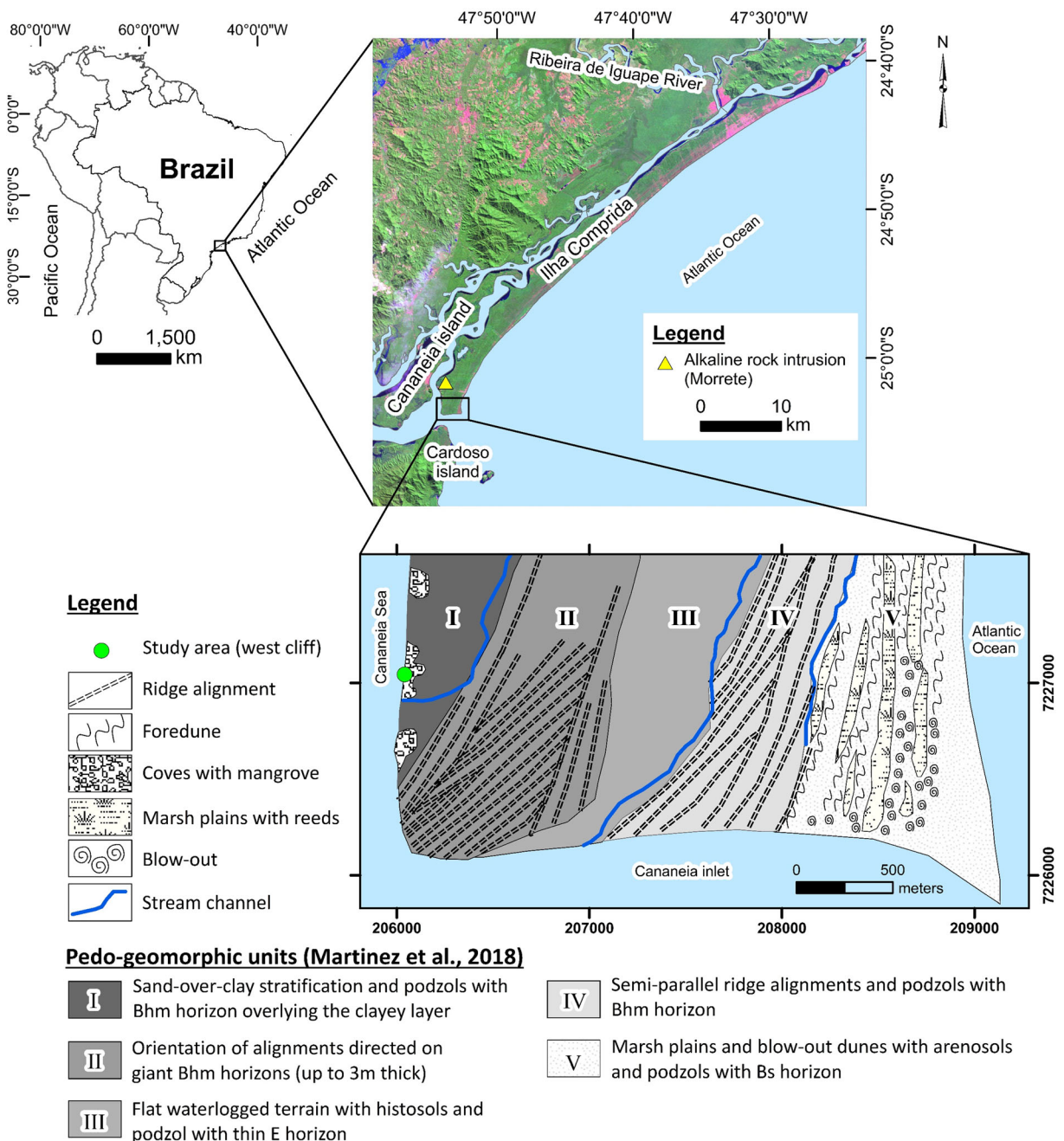


Fig. 1 Location of the study area in Ilha Comprida, southeastern Brazil

condition of the podzols close to the cliff for at least fifty years (Martinez et al. 2018).

The climate in this region is Tropical Wet (Af in Köppen's Classification System), with a mean annual precipitation of 2134 mm and a mean annual temperature of 23°C (Alvares et al. 2013). The coastal vegetation varies between grassy, heath, herbaceous swamp,

swampy forest, scrubby forest, and high-canopy rainforest (Souza and Luna 2008). The tropical rainforest in this coastal environment is an ecosystem called *restinga*, which is a marginal component of the Atlantic Forest. Restinga vegetation occurs just above sea level in tropical-subtropical humid conditions on coastal sandy soils with very low natural fertility. The

principal factors that influence this ecosystem are groundwater level, nutrient cycling, distance from the shore, a saline groundwater wedge at its borders, and sea-salt spray (Gomes et al. 2010). The decomposition of restinga litter and roots produces a large amount of DOM that moves to groundwater and streams, resulting in a water table that is enriched with DOM (Gomes et al. 2007; Coelho et al. 2010; Martinez et al. 2018). In our study area, podzols are the dominant soils in the restinga forest.

Many species of trees occur in restinga forests, including Cashew trees (*Anacardium occidentale*), which are native to a wide geographical area in Brazil, including the coastal plains, savannas (Cerrados), and the semi-arid region (Caatinga) of the northeastern states (Mitchell and Mori 1987; Luz 2011). The spread of *Anacardium occidentale* within the coastal plains has intensified as a result of planting in the region within the last 400 years principally due to economic interest in its products (fruit, nut, oils, etc.). Cashew trees are particularly tolerant of sandy and saline soils, which is why these trees are common components of coastal and restinga vegetation communities. Importantly, cashew trees have adopted several strategies for tolerating drought stress, for instance, the trees are deeply rooted, often forming a large taproot that can extend well beyond 4 m, with the roots representing ~ 20 % of the dry weight of semi-mature trees (Carr 2014).

Soil morphology description and sampling

Soil morphology was described in a South-North transect of 35 meters length located on the west-facing cliff (Fig. 2). The transect was separated into four compartments based on large-scale morphologic characteristics (Fig. 2a). While this study is principally focused on understanding the mechanism behind the abrupt disappearance of the Bh horizon on the left of profile PC1 located between compartments 2 and 3 (Fig. 2a), we rely on the interpretation of the morphology of the entire transect.

The following morphological attributes were described: horizon boundary (distinctness and topography), color, structure, texture, consistence, cementation, and roots (quantity and size). Our description used the terminology and standards from Schoeneberger et al. (2012). Profile PC1, where a tree-root was found at 135 cm depth, marks an abrupt lateral change in the morphology of the B horizon (Fig. 2b). This profile is

classified as an Albic Carbic Ortsteinic Podzol (Arenic, Oxyaquic) according to World Reference Base for Soil Resources (IUSS Working Group WRB 2015) and was sampled by pedogenetic horizons (Fig. 2b) for subsequent analysis and micromorphological description.

Physical and chemical analyses

Soil samples from 8 pedogenetic horizons of profile PC1 were air-dried and sieved (2 mm), then analyzed for particle size distribution (i.e. the percentage of sand (2 to 0.05 mm), silt (0.05 to 0.002 mm), and clay (≤ 0.002 mm)) and routine chemistry for classification purposes. Clay content was estimated using the hydrometer method, while the sand content was determined by sieving (Embrapa 2011). For each pedogenic horizon, we collected a soil sample at the midpoint depth of the horizon. The location of the soil samples is shown in Fig. 2b by the pedogenic horizon labels A, E, BhE, Bh, Bhm, 2Cgj, and 3 C. The percentages of sand and clay in the Bhm and 2Cgj horizons were used to estimate the saturated hydraulic conductivity based on a pedotransfer function ($K_{sat_pedotransfer}$, Eq. 1) developed by Cosby et al. (1984) and summarized by Zhang and Schaap (2019). We converted the $K_{sat_pedotransfer}$ value to m/s using the following conversion factor: 1 cm/day = 1.16×10^{-7} m/s.

$$K_{sat_pedotransfer}(\text{cm/day}) = 60.96 \times 10^{0.0126 \times \text{sand} - 0.0064 \times \text{clay} - 0.6} \quad (1)$$

Organic carbon content was determined by wet oxidation (Walkley-Black) using $\text{K}_2\text{Cr}_2\text{O}_7$ (0.4 mol L^{-1}) and back-titration with $(\text{NH}_4)_2\text{Fe}(\text{SO}_4)_2 \times 6\text{H}_2\text{O}$. Organically complexed Fe (Fe_p) and Al (Al_p) were extracted with sodium pyrophosphate $\text{Na}_4\text{P}_2\text{O}_7 \times 10\text{H}_2\text{O}$ (0.1 mol L^{-1}) at pH = 10 (McKeague 1967; DeConinck 1980; Embrapa 2011). The concentrations of extracted Fe and Al were measured with an atomic absorption spectrophotometer. Soil pH in water 1:2.5 (soil:water) was measured after shaking for 5 minutes.

Soil micromorphology

An undisturbed soil sample was collected from profile PC1 at 135 cm of depth in the area affected by roots from a cashew tree. This sample was used to prepare a thin section of the transition between the E horizon, the vertical OM-band, and the Bh horizon of profile PC1

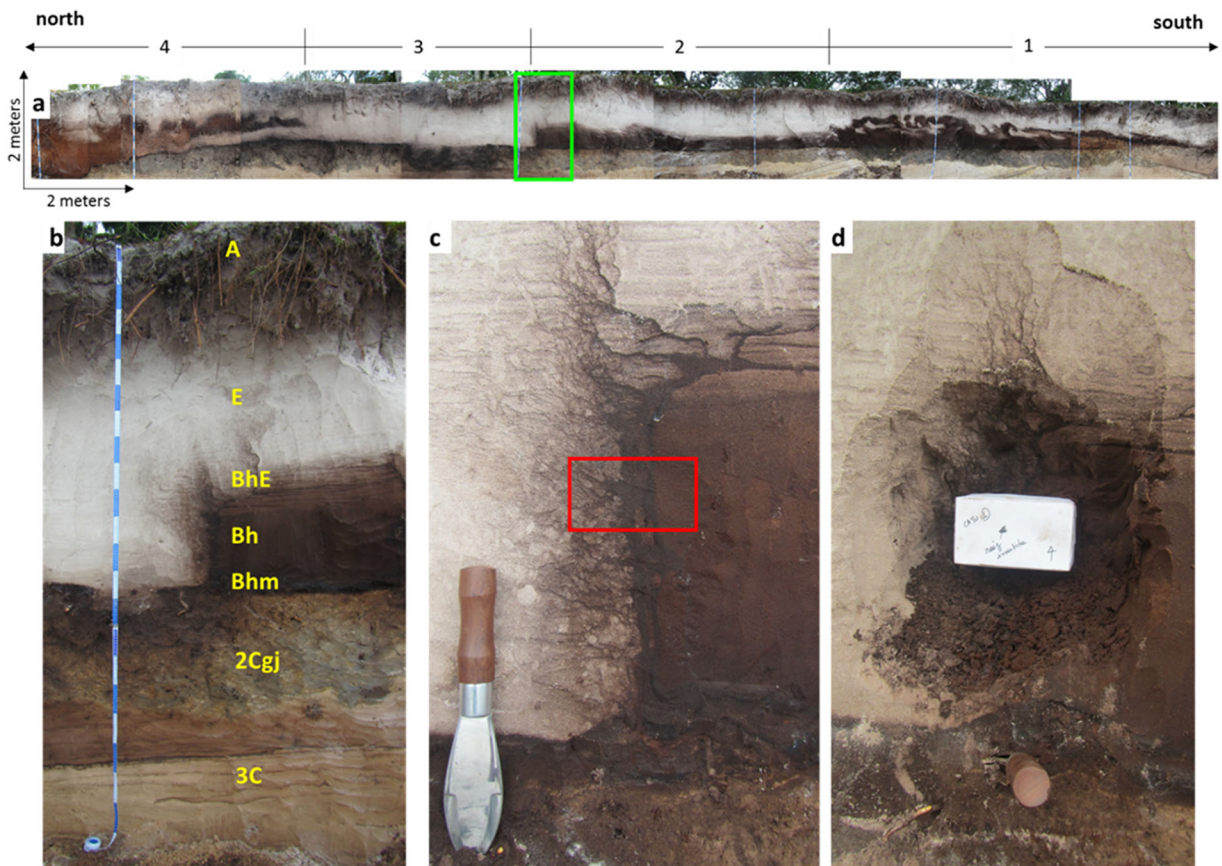


Fig. 2 **a** Location of profile PC1 (green frame) in the studied cliff. Soils were divided into compartments (1, 2, 3, and 4) according to morphological similarity. **b** Profile PC1 with irregular (step-like) boundary between the E and Bh horizons. The labels A, E, BhE, Bh, Bhm, 2Cgj, and 2 C indicate the pedogenic horizons of profile PC1 described in Tables 2 and 3. These codes also indicate where samples for physical and chemical soil analysis were taken from.

c Vertical OM-band following a large root of a cashew tree. Note the white rounded and elongated mottles due to the microbiological degradation of OM in the EB transition (vertical and lateral). The red frame indicates where the sample for thin sections was taken from. **d** Location of the sample collected for thin sectioning at the transition between the E and Bh horizons and which is defined by the vertical band of OM

(Fig. 2c and d). The soil sample was manually excavated and horizontally oriented during collection as a rectangular soil block using a cardboard box. The undisturbed soil block inside the cardboard box was then carefully wrapped with plastic film to prevent structure deformation. The preparation of thin sections followed the procedures of Murphy (1986), which are described as follows: (1) the soil block was completely air-dried; (2) a 1:1 solution of polyester resin and styrene monomer, to which we added fluorescent pigment (Tinopal OB) and a catalyst (Butanox) was prepared; (3) using the prepared solution, the dried soil block was impregnated using a vacuum system; (4) after complete hardening, the blocks were cut into slabs of $18 \times 70 \times 50$ mm and polished to obtain thin section with a thickness of 30 μ m. Finally, the thin sections were glued to 5×8 cm

glass slides for examination under a trinocular petrographic microscope coupled to a charge-coupled device (CCD) camera from the Optical Microscopy Laboratory of the Department of Soil Science ESALQ/USP (Piracicaba, Brazil). The micro-features were described according to the terminology of Bullock et al. (1985) and OM forms according to DeConinck et al. (1974).

Image processing, soil porosity, and hydrologic parameters

Soil porosity, permeability, capillary pressure, and saturated hydrologic conductivity (K_{sat}) were estimated for the E horizon, the vertical OM-band, and the Bh horizon of profile PC1 (Fig. 2c and d). To this end, the thin section was illuminated under ultraviolet light to

enhance the color contrast between pores and solids. Then, 3 regions each from the E-horizon, OM-band, and Bh horizon were photographed with the CCD image sensor coupled with the microscope. The regions are from the 3 studied positions (E-horizon, OM-band, and Bh-horizon). The selected regions have dimensions of $3220 \times 2415 \mu\text{m}$ with a resolution of 644×483 pixels, thus, the area of one pixel is $5 \times 5 \mu\text{m}^2$.

Three image processing steps were done for all 9 selected regions. First, we opened the images in ImageJ and set the scale according to the original scale bar in the photos from the CCD image sensor coupled with the microscope. Second, we generated the binary images according to the following stepwise procedure in ImageJ: Image $\rightarrow$ Adjust $\rightarrow$ Color Threshold $\rightarrow$ Select $\rightarrow$ Process $\rightarrow$ Binary $\rightarrow$ Make Binary. The binary images have reliable information of soil porosity because they are from the photos of the thin section taken under ultraviolet light which increases the color contrast between pores and grains (Cooper et al. 2016). Third, the color contrast allows the separation of pores and solids using the binary images that were treated according to the following stepwise procedure in ImageJ: Plugin $\rightarrow$ BoneJ $\rightarrow$ Analyze $\rightarrow$ Particle Analyser. The command “Particle Analyser” gives the “Result” table that shows the data for surface area of the pores, as well as the feret pore diameter and minor radius. The feret diameter consists in the distance between the two parallel planes restricting the largest cross-section of the pore. The minor radius is the length of best-fit ellipsoid’s short radius of the cross-section of the pores, which can be assumed as the critical pore size for permeability (Nishiyama and Yokoyama 2017).

Porosity (ϕ) expressed in % was determined as the sum of the surface area of the pores in relation to the total area of the image (Eq. 2).

$$\text{Porosity } \phi(\%) = \frac{\sum \text{surface area of the pores} \times 100}{\text{total area of the image}} \quad (2)$$

The minor radius of the pores (r) obtained with BoneJ plugin in ImageJ was converted from μm to m. Then, we estimated the permeability (k) expressed in m^2 according to Eq. 3 (Nishiyama and Yokoyama 2017), where ϕ is the porosity (%), r is the minor pore radius (m), and c is the geometric factor or constant that we kept equals 1.

$$\text{Permeability } k (\text{m}^2) = c \times \phi \times r^2 \quad (3)$$

Saturated hydraulic conductivity based on image processing (Ksat) expressed in m/s was estimated according to Eq. 4 (Fryar and Mukherjee 2021), where k is the permeability calculated with Eq. 3, ρ is the density of water (1000 kg/m^3), g is gravity (9.8 m/s^2), and μ is the dynamic viscosity of water (0.0016735 Pa/s).

$$k_{\text{sat}} = (m/s) = \frac{k \times \rho \times g}{\mu} \quad (4)$$

Capillary pressure (h) expressed in cm was estimated according to the Young-Laplace equation (Eq. 5), where γ is the surface energy (72.7 g/s^2), β is the contact angle of the water which we kept constant equals 1, g is the gravity (980 g/cm^2), ρ is the density of water (1 g/cm^3), and R corresponds to $\frac{1}{2}$ of the average feret diameter obtained from image processing with BoneJ plugin.

$$\text{Capillary pressure } h (\text{cm}) = \frac{-2\gamma \cos \beta}{\rho g R} \quad (5)$$

The statistical differences between the studied positions (Bh horizon, OM-band, and E-horizon) in terms of porosity, permeability, Ksat, and capillary pressure were established with the two-way ANOVA and Tukey’s HSB tests in RStudio software.

Dating by OSL and ^{14}C

OSL ages provide the minimum age for the sediment deposition and the maximum for podzol formation. We determined OSL ages of profile PC1 on five samples taken from the following soil depths: 35 cm (E horizon); 95 cm (BhE horizon); 110 and 130 cm (Bh horizon); and 200 cm (3 C horizon). The horizontal samples were collected without exposure to light, using opaque PVC pipes (30 cm long and 5 cm in diameter) that were introduced into the soil horizons perpendicular to the cliff. OSL determinations were carried out at the Laboratory of Datação Ltda (Brazil) applying five quartz aliquots and the single aliquot regeneration-dose (SAR) protocol (Wallinga et al. 2000). All laboratory pre-treatments were conducted under red light, as follows. First, we separated the 120–150 μm soil materials by sieving. Second, the material retained on the sieve was treated several times with 20 % H_2O_2 , 20 % HF, and 10 % HCl (40 minutes) to remove OM, CaCO_3 , feldspars, and the outer parts of quartz grains affected by

alpha radiation. Density separation in lithium metatungstate was performed to separate heavy ($> 2.75 \text{ g/cm}^3$) from light minerals and quartz from feldspar ($> 2.62 \text{ g/cm}^3$). Moisture content and the concentrations of isotopes $^{238}\text{U}+^{235}\text{U}$, ^{232}Th , and ^{40}K were determined for the estimation of radiation dose-rate. These data are presented in Table 2, which also contains sampling depth, annual dose ($\mu\text{Gy/year}$), mean paleodose (Gy), and age with respective errors for the five quartz aliquots.

Two ^{14}C age determinations of roots embedded in the Bh and 2Cgj horizons were carried out. The ^{14}C age of a root embedded in the clayey 2Cgj horizon is used to constrain the minimum age for the overlying sediment. A large cashew root in the transition between the E and Bh horizons was used to determine the age of the tree and thus the timescale associated with morphological alteration of the podzol morphology. Prior to analysis, both roots were rinsed in hot HCl to eliminate carbonates. To remove secondary organic materials, the samples were then treated with NaOH. The determinations of ^{14}C ages were performed by Beta Analytic Inc. (Miami, FL, USA) using Accelerator Mass Spectrometry (AMS). The ages were calibrated in the 2-sigma range before the present (cal. yrs BP) using the curve for the southern hemisphere with a 95 % probability (Hogg et al. 2013).

Results

Soil macromorphology

Observing the whole transect, we identified four compartments of morphologically similar soils (Fig. 2a). Compartment 1 is in the southern part of the transect where podzols have irregular and broken boundaries between the Bh and E horizons (Fig. 3a). We found parts of the E horizon inside the Bh horizon (Fig. 3b), as well as convoluted, ‘wave-shaped’ EB horizon transitions (Fig. 3c). In compartment 2, the boundary between the E and B horizons has a flat topography which is characteristic of poorly drained podzols formed by a lateral flux of DOM in the perched water table on the 2Cgj horizon (Fig. 4a). The 2Cgj horizon has gley color (Gley1 5/N) with Fe-mottles (10YR 6/8), while the Bhm horizon (5YR 2.5/1) has a darker color and stronger cementation than the Bh horizon (5YR 3/2).

The transition between compartments 2 and 3 is defined by an abrupt disappearance of the Bh horizon, which causes a step-like transition between the E and B horizon (Fig. 2b). Profile PC1 is located at the center of the transect where the soil is up to 300 cm thick and exhibits the pedogenic horizons A, E, BhE, Bh, Bhm, 2Cgj, 3 C, and 4 C (Table 1; Fig. 5a). The descriptions in Tables 1 and 3 refer to the sequence of pedogenic horizons of profile PC1 shown in the right part of Fig. 2b. The boundary distinctness of the subsurface horizons ranges from abrupt to clear. These marked horizon boundaries reflect the presence of a clayey 2Cgj horizon (color Gley1 5/N) between sand materials with distinct colors, 5YR 3/2 and 5YR 2.5/1 of the B horizons, 7.5YR 7/2 of the 3 C horizon, and 5YR 4/1 of the E horizon. This transition between the E and Bh horizon of profile PC1 is marked by a vertical OM-band that surrounds a large cashew taproot (Fig. 5b). The OM-band and the large taproot separate compartments 2 and 3.

Compartment 3 starts on the left part of Fig. 2b where the E horizon extends much deeper into the soil (130 cm depth) than in profile PC1 with the E horizon ending at 90 cm depth (Table 1). The disappearance of the Bh horizon in compartment 3 results in a thicker E horizon (110 cm thick) compared to the other compartments (Fig. 4b). Also, the 2Cgj horizon in compartment 3 is darker than in the other compartments. Small and medium size roots branch laterally from the taproot in profile PC1 towards compartment 3 (Figs. 4b and 5b).

In profile PC1, cashew roots are dominated by a deep taproot 15 cm in diameter that extends vertically down to 200 cm depth in the profile (Fig. 5b and Supplementary Data 1). This taproot has lateral branches that markedly changed the appearance of soil horizons, resulting in wedge or step-like features. Effects of large tree roots are not obvious in compartment 4. Here, the primary pedogenic process appears to be the extension of the E horizon under the Bh horizon interpreted to be a result of lateral flow of DOM-poor water being concentrated at the contact with the 2Cgj horizon (Fig. 4c). The increased flux of water resulted in the E horizon extending laterally into what was the bottom of the Bh horizon and which is enhanced by the topographic slope inclined towards the north and east in this part of the transect. However, while the bottom of the Bh horizon appears to be

Fig. 3 **a** Compartment 1 where tree roots appear to have influenced the formation of irregular and broken boundaries between the E and B horizons. The codes A, E, BhE, Bh, Bhm, 2Cgj, and 3C indicate soil horizons. **b** and **c** Detail of the BhE horizon transition showing the extension of the E horizon into the Bh horizon around tree-roots (arrows)

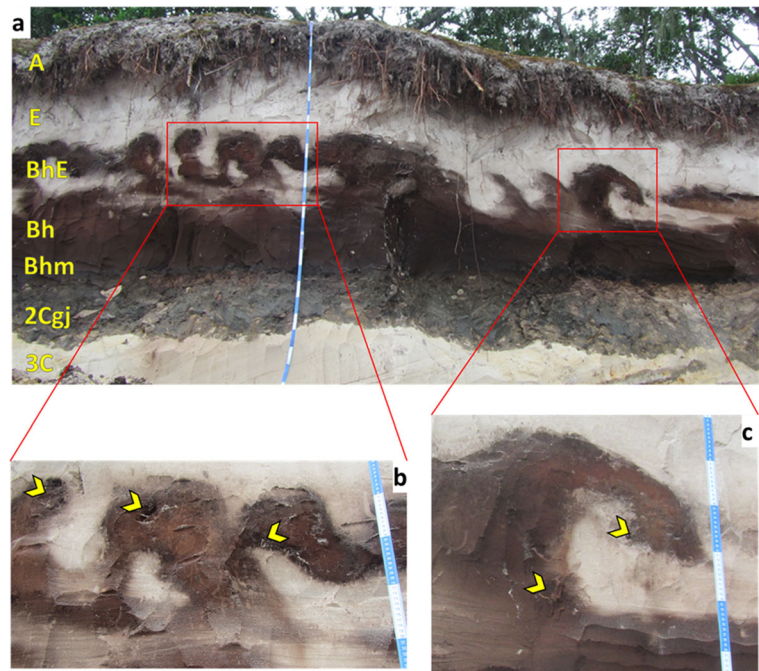


Fig. 4 **a** Compartment 2 with the flat transition between the E and B horizons without tree-root. DOM is immobilized in the Bhm horizon (cemented) and in the Bh horizon because the impervious clayey 2Cgj horizon precludes vertical water flow.

b Compartment 3 lacks the B horizon, while the 2Cgj horizon (yellow arrow) is darker and richer in OM than in the other compartments. **c** The E horizon laterally enters the bottom of the B horizon due to enhanced lateral water flow (yellow arrow) over the 2Cgj horizon. The red arrows in the uppermost Bh horizon indicate the presence of roots in different decomposition stages that (i) add OM and contribute to the formation of the Bh horizon, and (ii) create paths of water flow where “tongues” of the E horizon into the B horizon are formed

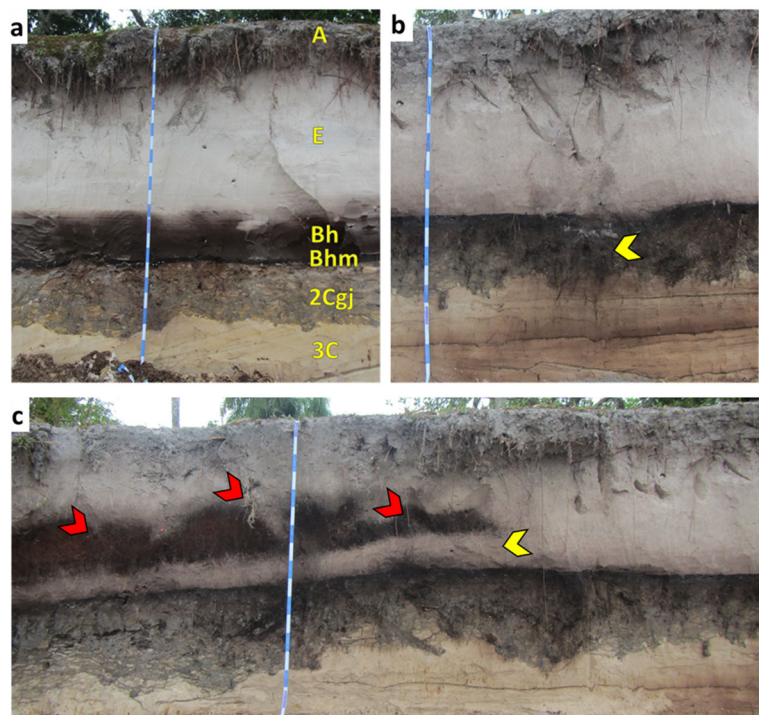


Table 1 Morphological attributes of soil profile PC1 classified as Albic Carbic Ortsteinic Podzol (Arenic, Oxyaquic)

Hor. ¹	Depth (cm)	Hor. boundary		Str. ⁴	Tex. ⁵	Color	Cons. ⁶	Cement. ⁷	Roots	
		Distin. ²	Topo. ³						Quant. ⁸	Size ⁹
A	0–23	gradual	smooth	SGR	S	5YR 5/1	loose	none	many	VF/F/M/C
E	23–90	clear	wavy	SGR	S	5YR 8/2	loose	none	few	F/M
BhE	90–105	clear	irregular	SGR	S	5YR 6/1	friable	none	few	F/M
Bh†	105–132	abrupt	irregular	SGR	S	5YR 3/2	firm	moderated	common	M/C/VC
Bhm†	132–140	abrupt	smooth	SGR	S	5YR 2.5/1	very firm	strong	common	F/M/C/VC
2Cgj†	140–210	abrupt	wavy	MA	SC/C	Gley1 5/N, 10YR 6/8	firm	none	few	M/C/VC
3 C	210–225	clear	smooth	MA	S	7.5YR 7/2	loose	none	none	none
4 C	225–300+			SRG	S	7.5YR 8/2	loose	none	none	none

¹ Soil horizon; ² Horizon Distinctness; ³ Horizon Topography; ⁴ Soil Structure (SGR: single grain; MA: massive); ⁵ Soil texture determined in the field (S: Sand, SC: Sandy Clay, C: Clay); ⁶ Soil consistence; ⁷ Soil cementation; ⁸ Roots quantity; ⁹ Roots size (VF: very fine, F: fine, M: medium, C: coarse, VC: very coarse).

† Presence of large roots of a cashew-tree (*Anacardium occidentale*).

undergoing degradation, we found fine roots at different stages of decomposition which ostensibly added OM to the upper part of the Bh horizon (Fig. 4c).

OSL and ¹⁴C determinations

The calibrated ¹⁴C age (6495 – 6400 cal. yrs BP) of the root remnant embedded in the clayey matrix, indicates that the deposition of the sediment of the 2Cgj horizon occurred around 6000 years ago. This ¹⁴C age closely matches the OSL age (6041 ± 503 years) obtained from the sandy sedimentary facies outside of our study area (Guedes et al. 2011), thus confirming a basal age of ~ 6000 years ago. Additionally, OSL dosages measured in this study indicate a consistent vertical decrease in burial age from the bottom of profile PC1 (3 C horizon with 5380 ± 760 years) to the E horizon (3160 ± 370 years) (Table 2). These OSL data are similar to the sedimentation ages of Ilha Comprida (3240 ± 510 and 3570 ± 590; Souza and Perez Filho 2019). Considering these age determinations of the parent material of this podzol, we infer that podzolisation cannot have started before 4000 years ago.

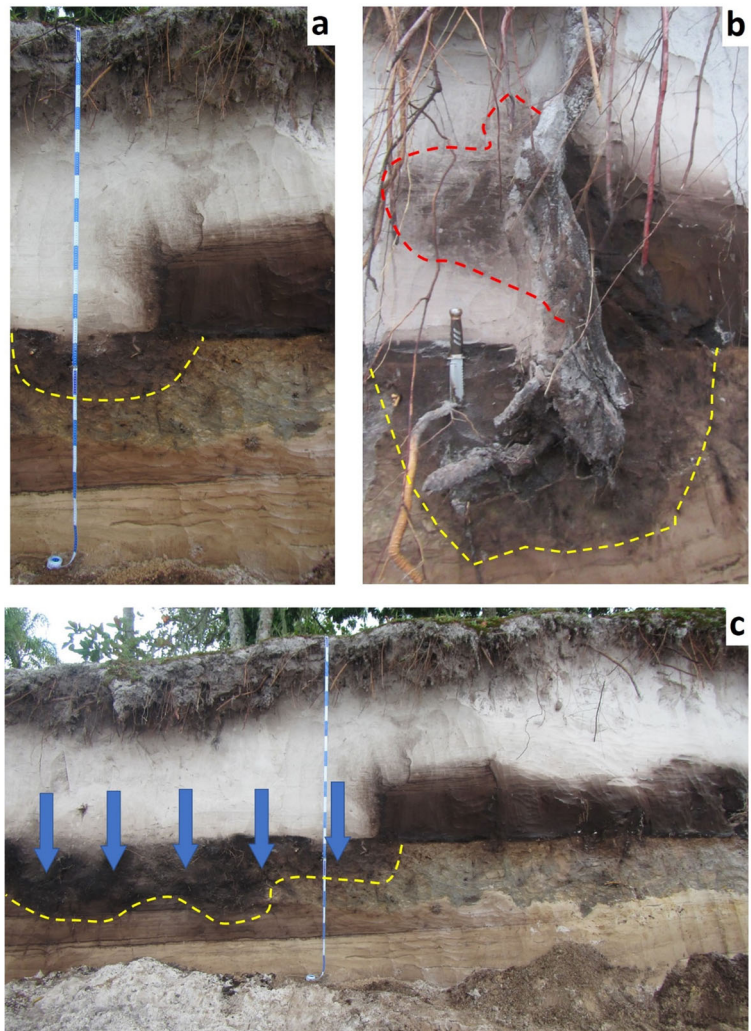
The root that defines the lateral boundary between the E and B horizon of profile PC1 (Fig. 5b) has a modern conventional ¹⁴C age (118.2 ± 0.3 pMC – percent modern carbon) which indicates that it is from after 1950. Considering that this tree-root may have caused the remarkable modifications in the morphology of PC1 profile (e.g., total degradation of the Bh horizon, Fig. 5),

we infer that, in areas with tropical coastal rainforest, the morphology of a podzol may substantially change within a short time (~ 50 years).

Physical and chemical soil properties

Except for the 2Cgj horizon that has 36 % clay, all other soil horizons of profile PC1 have, on average, ≥ 97 % sand (Table 1). The particle size distribution determined in the lab corroborated our texture estimations in the field. Based on the contents of sand and clay, we estimate $K_{sat_{pedotransfer}}$ for the Bhm and 2Cgj horizons using the pedotransfer function developed by Cosby et al. (1984). Our estimations indicate that the 2Cgj horizon has significantly lower $K_{sat_{pedotransfer}}$ (6.3×10^{-6} m/s) than the Bhm horizon (3.0×10^{-5} m/s). However, the Bhm horizon is strongly cemented which should constrain the penetration from the overlying Bh horizon that is not cemented. Likewise, the massive non-aggregate structureless of the 2Cgj horizon should reduce the vertical water flow that percolates through the single grain material of the B horizons. Therefore, our $K_{sat_{pedotransfer}}$ values for the Bhm and 2Cgj horizons are likely overestimated and do not reflect the real situation. Although it is unknown which horizon is the most impervious layer in this profile, we assume that both, the Bhm and 2Cgj horizons, may limit the vertical water flow and root penetration. The formation of the 2Cgj horizon occurred prior to vegetation development and associated DOM inputs and podzolisation. It created an impervious layer that caused a perched water table

Fig. 5 **a** Profile PC1 before the exposure of the tree-root. The yellow dashed line indicates the lowermost boundary of a zone where organic matter entered the clayey 2Cgj horizon. **b** The tree-root was exposed after the erosion of the cliff. The red dashed line indicates the Bh horizon inboard the tree-roots whereas the yellow dashed lines show the 2Cgj horizon affected by root penetration. **c** Increased vertical flow around tree roots caused degradation of the B horizon and infiltration of DOM and OM of decaying roots into the clay layer (blue arrows)



before the formation of the Bhm horizon. With the development of the restinga forest, the perched water table has been loaded with DOM. The stagnation of DOM above the clayey layer was responsible for the

formation of the Bhm horizon that is enriched with OM ($C\%=2.4$). Below the clay layer, we found a sequence of 3 C and 4 C horizons that were again sandy with $\geq 95\%$ sand (Table 1).

Table 2 OSL ages and related sample characteristics for selected soil horizons of the west cliff

Horizon	Depth (cm)	Age (years)	Water content (%)	Th (ppm)	U	K (%)	Annual dose ($\mu\text{Gy}/\text{year}$)	Mean paleo-dose (Gy)/SD
E	35	3160 ± 370	2.9	3.3 ± 0.1	0.9 ± 0.1	0.3 ± 0.1	1050 ± 70	3.3 ± 0.1
BhE	95	3390 ± 460	3.7	7.9 ± 0.3	2.8 ± 0.3	0.6 ± 0.1	2160 ± 185	7.3 ± 0.3
Bh	110	3410 ± 530	4.1	3.3 ± 0.1	1.4 ± 0.2	0.4 ± 0.1	1260 ± 135	4.3 ± 0.1
Bh	130	3810 ± 730	4.8	3.3 ± 0.1	1.1 ± 0.3	0.6 ± 0.1	1340 ± 190	5.1 ± 0.2
3 C	200	4380 ± 760	9.3	3.0 ± 0.1	0.3 ± 0.1	0.5 ± 0.1	1000 ± 125	4.4 ± 0.1

Table 3 Particle size distribution and basic chemical properties of profile PC1

Horizon	Depth (cm)	Sand %	Silt	Clay	pH (H ₂ O)	C org. %	Fe _p	Al _p	C/Al _p
A	0–23	97	1	2	5.1	2.3	0.0	0.0	-
E	23–90	98	1	1	5.3	0	0.0	0.0	-
BhE	90–105	98	1	1	4.9	0.6	0.0	0.1	6.0
Bh†	105–132	99	0	1	4.5	1.5	0.0	0.3	5.0
Bhm†	132–140	98	1	1	4.5	2.4	0.0	0.4	1.7
2Cgj†	140–210	61	3	36	3.8	0.8	0.4	1.9	0.4
3 C	210–225	95	2	3	4.5	0.2	1.5	1.4	0.1
4 C	225–300+	98	1	1	5.0	0.0	0.1	0.5	0.0

† Presence of large roots of a cashew-tree (*Anacardium occidentale*)

Fe_p: Iron extracted with sodium pyrophosphate

Al_p: Aluminum extracted with sodium pyrophosphate

Organic C and Al_p contents increase in the Bh horizon (C%=1.5 and Al_p=0.3 %) relative to the overlying horizons (Table 1). The absence of Fe_p in the B horizons is an indicator that this podzol was formed under poor drainage and iron was reduced and lost as Fe²⁺. pH values are low in the B horizons (pH = 4.5 to 4.9) and even lower in the 2Cgj horizon (pH = 3.8) (Table 3). The low pH value and the high Fe content in the 2Cgj horizon are likely due to the oxidation of pyrite that was found in clayey horizons in the lagoons of this region (Gomes et al. 2007).

Soil micromorphology

The following description refers to the soil thin sections of the contact between the Bh horizon, the vertical OM-band, and the E-horizon in profile PC1 (Figs. 2d and 6a). The vertical OM-band has a gelfuric fine and coarse distribution. The fine material in this vertical feature consists mainly of monomorphic OM coatings (Fig. 6d) that caused cementation and formed the OM-band. In the Bh horizon, the OM coatings were thinner than in the OM-band (Fig. 6e and f). Monomorphic OM coatings in the B-horizon do not entirely fill the pores, which explains why this Bh horizon is not cemented. Although monomorphic coatings dominated the OM-band and the B horizon, we also found polymorphic OM pellets related to decaying roots (Fig. 6). As a result, the OM in the B horizon and OM-band is from DOM and decayed roots.

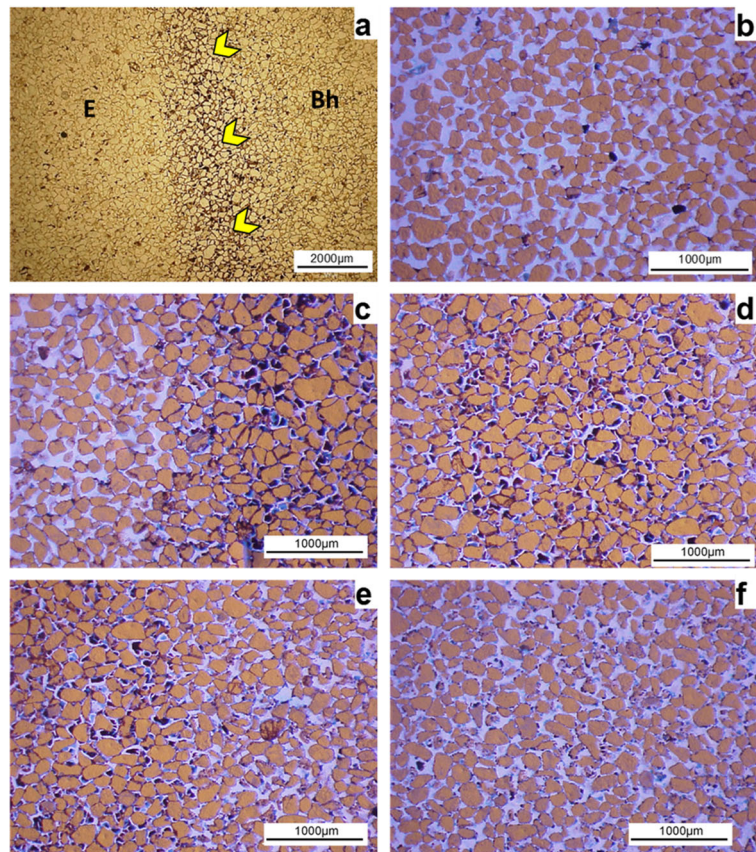
Because the monomorphic coatings in the OM-band and in the Bh horizon are not translucent, we infer that

some of these coatings are features of degradation of pellets and not of DOM, which agrees with a large contribution from decayed roots. This observation is supported by the high frequency of roots in this profile (Table 1; Fig. 2b and d). In contrast, the E-horizon to the left of the OM-band does not contain coatings (Fig. 6b) and has a monic distribution of f/c materials. The B horizon was likely continuous from compartments 2 to 4 (Fig. 2a) before the insertion of the tree-root. The improved drainage caused by the opening of preferential water paths upon root decay caused microbial degradation of OM coatings and converted the Bh horizon into an E horizon.

Soil porosity and hydrologic parameters

The study of thin sections under the microscope indicates that the arrangement of pores and grains differs between the Bh horizon, the OM-band, and the E horizon (Fig. 6). Sand grains are closer to each other in the OM-band compared to the E and Bh horizons (Fig. 6 and Supplementary Data 1). This resulted in a porosity that is significantly lower in the OM-band (42 ± 4 %) than in the E horizon (65 ± 3 %) and Bh horizon (61 ± 1 %), p-value < 0.001 (Table 4; Fig. 7a). Likewise, the average feret diameter of the pores is smaller in the OM-band (96 ± 12 μm) than in the E horizon (155 ± 11 μm) and Bh horizon (121 ± 8 μm), p-value = 0.004 (Table 4; Fig. 7b). Due to these significant differences, capillary pressure is stronger in the OM-band (-17 ± 2 cm) than in the E-horizon (-10 ± 1 cm) and the Bh horizon (-13 ± 1 cm), p-value = 0.012 (Table 4; Fig. 7c). Our

Fig. 6 Thin sections of the contact between the E horizon, the OM-band, and the Bh horizon in profile PC1. **a** Vertical OM-band (yellow arrows) at the boundary between the E and Bh horizons. **b** Large porosity of the E horizon. **c** Contact between the E horizon and the vertical OM-band. Sand grains are closer to each other in the OM-band than in the E horizon. **d** Detail of the OM-band with low porosity due to the growth of a cashew root. In the OM-band, there is polymorphic OM derived from root decomposition, but also monomorphic OM brought by DOM from the groundwater. **e** Contact between the vertical OM-band and the Bh horizon. Sand grains in the OM-band are more closely packed than in the Bh horizon. **f** Bh horizon with polymorphic OM derived from root decomposition



estimations for permeability are significantly higher in the E and Bh horizons ($\geq 8.9 \times 10^{-10} \pm 5.4 \times 10^{-11} \text{ m}^2$) compared to the OM-band ($6.6 \times 10^{-10} \pm 8.0 \times 10^{-11} \text{ m}^2$), $p\text{-value} = 0.007$ (Table 4; Fig. 7d). As a result, the E and Bh horizons have also a greater K_{sat} ($\geq 5.2 \times 10^{-4} \pm 3.2 \times 10^{-4} \text{ m/s}$) than the OM-band ($3.8 \times 10^{-3} \pm 4.7 \times 10^{-4} \text{ m/s}$), $p\text{-value} = 0.007$ (Table 4; Fig. 7d). The porosity contrast due to tree-root growth created a capillary break (Khire et al. 2000) which resulted in the accumulation of DOM in a vertical OM band.

Discussion

Drainage evolution affecting podzol morphology

Podzols in this study were formed under poor drainage conditions because the sandy Bh horizon overlies the cemented Bhm horizon and an impervious clayey 2Cg_j horizon that created a perched water table bearing DOM. Due to erosion of the cliff, the drainage of these soils was improved (Martinez et al. 2018). Since

precipitation is high (2134 mm annually), transport of DOM through the preferential water paths created by tree-root decay is enhanced. As DOM passes through the soil, it removes complexed metals (e.g., Al) from its surroundings. Subsequently, OM at the top of the B horizon is removed through microbial oxidation (Buurman et al. 2005) as we observed in compartments 1 and 4 (Figs. 3 and 4c). Therefore, with improved drainage and removal of complexed metals in areas of preferential water flow, there was an increased microbial decomposition of OM (Buurman et al. 2005; Lopes-Mazzetto et al. 2018). Owing to the improved drainage, the top of the B horizon degraded and E horizon tongues penetrated the original poorly drained Bh horizon (Lopes-Mazzetto et al. 2018). The degradation of OM along preferential water paths is responsible for the irregular or broken boundaries between the Bh and E horizons. Similar lateral variability in podzol morphology controlled by parent material stratification and drainage condition were identified in other parts of the coastal plain of Sao Paulo State (Gomes et al. 2007; Coelho et al. 2010). Here, we inferred that with

Table 4 Porosity and hydrological parameters of the area affected by the tree-root where a vertical OM-band was formed at 135 cm of depth in profile PC1

	Averages for the studied position in the soil thin section		
	E-horizon (n=3)	OM-band (n=3)	Bh-horizon (n=3)
Porosity (%)	65±3	42±4	61±1
Feret pore diameter (μm) ^a	155±11	96±12	121±8
Capillary pressure (cm) ^b	-10±1	-17±2	-13±1
Permeability (m ²) ^c	$9.6 \times 10^{-10} \pm 5.2 \times 10^{-11}$	$6.6 \times 10^{-10} \pm 8.0 \times 10^{-11}$	$8.9 \times 10^{-10} \pm 5.4 \times 10^{-11}$
Ksat (m/s)	$5.6 \times 10^{-3} \pm 3.1 \times 10^{-4}$	$3.8 \times 10^{-3} \pm 4.7 \times 10^{-4}$	$5.2 \times 10^{-3} \pm 3.2 \times 10^{-4}$

^a Feret diameter is the distance between the two parallel planes restricting the largest cross-section of the pore

^b Based on Young-Laplace equation and considering that the average pore radius equals to ½ of feret pore diameter

^c Based on the equation from Nishiyama and Yokohama (2017) and the critical pore size being the average minor radius of the pores, i.e. the length of best-fit ellipsoid's short radius of the cross-section of the pores

improved drainage, the roots of cashew trees grew deeper than when the soil was poorly drained (Fig. 8).

Podzol morphology altered by root-mediated hydrologic processes

Interfingering of the E horizon and the B horizon is commonly interpreted as evidence of a top-down degradation of the Bh horizon due to improved drainage (Lopes-Mazzeto et al. 2018). However, improved drainage can be caused by decayed tree roots that create preferential, especially through the underlying aquitards (Bhm and 2Cgj horizons). In the present case, the tree-root penetrated the cemented Bhm horizon and the massive clayey 2Cgj horizon and thus increased the vertical water flow and the displacement of the OM from the B horizon to part of the 2Cgj horizon (Figs. 5 and 8). As a result, the 2Cgj horizon became darker and richer in OM than in the other compartments (Fig. 5c). Thus, the podzolisation process surpassed the physical barrier. Owing to the removal of OM from a former Bh horizon, we infer that the E horizon had a downward expansion (1 m thick) until it reached the cemented Bhm horizon (Fig. 8), which possibly remained water-saturated. Many studies have also found that improved drainage is localized in parts of the soil affected by rooting and burrowing (Noguchi et al. 1999; Uchida et al. 2002, 2005; Lin 2010; Buurman et al. 2013; Martinez et al. 2019).

In profile PC1, we observed that the B horizon reappeared landward, which indicates that the effect of the root on hydrology is local (Fig. 5b) and extends less

than 50 cm from the root. Where the tree-root penetrated the cemented Bhm horizon and the impervious clayey layer (2Cgj horizon), we observed that this disruption triggered an enhanced vertical water flow that promoted the degradation of the Bh horizon far distant from the tree-root. The preferential flow along the large cashew roots can be classified as *pipe flow* (Uchida et al. 2005; Johnson and Lehmann 2006; Li et al. 2018). Pipe flow has dual effects on the podzol. During wet periods, increased flows of water with DOM removes metals from organic complexes in the surroundings. During subsequent dry periods, the decomposition of the decomplexed OM increases. This process is responsible for the removal of OM from the Bh horizon in compartment 3. Over time, the white E horizon volumes inside the Bh horizon coalesce (Fig. 4c). The degradation of the Bh horizon along preferential water paths opened by tree-roots may be enhanced by uprooting (e.g., in compartment 1), and can be responsible for the formation of convoluted, wavy features, and rounded BhE horizon volumes (Fig. 3). As large trees fell towards the edge of the eroded cliff, the uprooting process may promote the deformation of the subsoil. Under this scenario, the taproots of falling trees essentially dragged the B horizon into the E and vice versa (Fig. 3c). Similar effects of complete soil profile inversion by tree uprooting which caused dramatic change in podzol morphology were observed in temperate forests (Schaetzl 1986; Shouse and Phillips 2016; Pawlik and Šamonil 2018).

We also observed that the growth of a taproot compressed the soil matrix surrounding the root, which rearranged sand grains in profile PC1 creating a zone

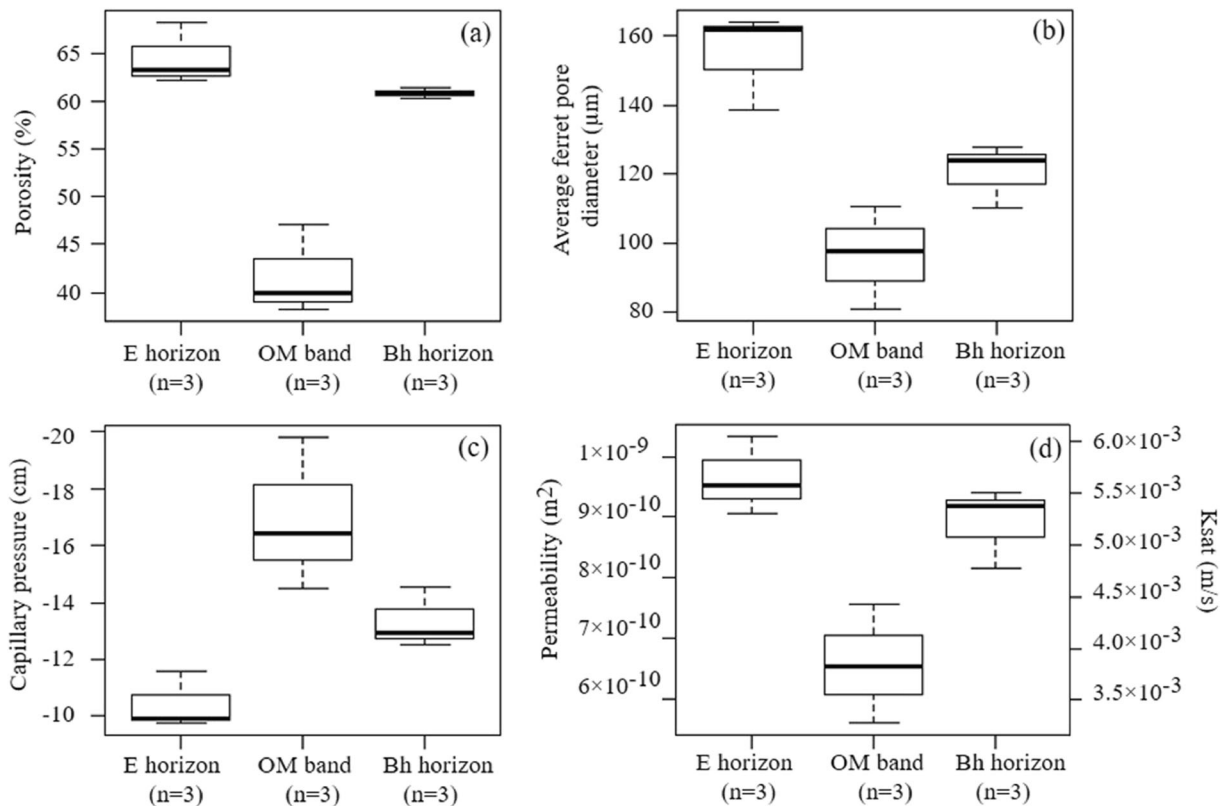


Fig. 7 Porosity (a), average feret pore diameter (b), permeability, and saturated hydraulic conductivity (Ksat) (d) are significantly lower in the vertical OM-band than in the adjacent E and Bh horizons. Capillary pressure (c) is stronger in the OM-band than in the E and Bh horizons. Feret diameter is the distance between the two parallel planes restricting the largest cross-section of the pore. Capillary pressure is based on Young-Laplace equation where pore radius was considered equals to $\frac{1}{2}$ of feret pore diameter. Permeability is based on the equation from Nishiyama and Yokohama (2017) and the critical pore size being the average minor radius of the pores, i.e. the length of best-fit ellipsoid's short radius of the cross-section of the pores. Variations in porosity and hydrologic parameters are associated with the growth of the

taproot (15 cm diameter and 200 cm length) of a cashew-tree (*Anacardium occidentale*). Water containing DOM is retained at a greater strength in the band than in the E and Bh horizons. The data range is represented by the boxes, whereas the bold horizontal line inside the boxes represents the mean. The 25th percentiles above and below the range are indicated by the dashed lines with a bar outside the boxes. The data in the graphs are based on the dimensions of the cross-section of the pores that were quantified from 3 binary images (Supplementary Data 1) of each studied position (E horizon, OM-band, and Bh horizon) in the soil thin section. Image processing and data collection were carried out using Image J software and BoneJ plugin

of lower porosity where DOM was retained, leading to the formation of a cemented vertical OM-band. This rearrangement of sand grains was recognized in the soil thin section where the OM-band shows less internal porosity (Figs. 6 and 7a) and greater capillary pressure than the E and Bh horizons (Fig. 7c). In fact, the OM-band has smaller average pore diameter and significantly lower permeability and Ksat than the adjacent B and E horizons (Fig. 7). The growth of the tree-root pushed the sand grains close to each other, decreasing the porosity of the area affected by the root. Therefore, our results are consistent with previous studies that observed

that the growth and death of large roots alter local porosity (Brantley et al. 2017; Sonnenborg et al. 2017). Another influence of tree-roots on the accumulation of OM in the studied vertical band is root mortality and decay adding OM in the B horizon (Hodge et al. 2009). The paths of preferential water flow opened by decaying tree-roots enable a stronger transport of DOM into the subsoil which increases the input of OM in parts of the B horizon. Such a contribution of tree-roots to podzolisation was also observed in a temperate climate (Pawlik and Šamonil 2018). Moreover, roots can function as a pump for selective absorption of water and

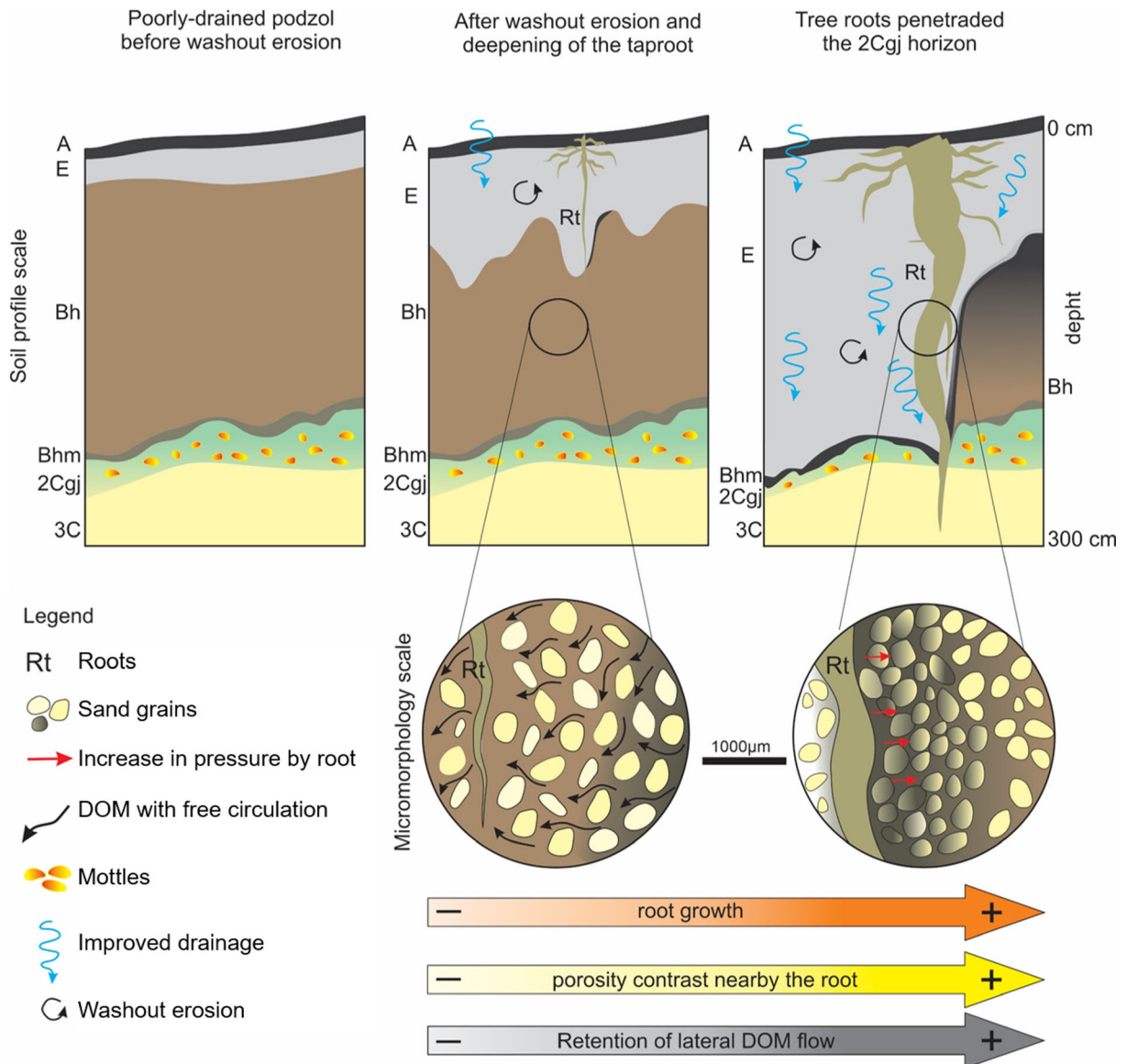


Fig. 8 Schematic representation of the processes associated to the modification of soil morphology of profile PC1 at different scales. Within 60 years, the tree-root grew and pushed the sand grains close to each other, increasing the porosity contrast in the vicinity of the root. The formation of a vertical OM-band was triggered by

both the circulation of DOM and the pump activity of the roots, leaving behind high contents of OM. As the taproot penetrated and ruptured the impervious 2Cgj horizon, the local drainage the left of the taproot was improved, which increased the vertical water flow responsible for the removal of OM from the Bh horizon

nutrients, leaving behind high contents of OM in the rooting zone in podzol Bh-horizon (Coelho et al. 2012).

In summary, taproots (e.g., from cashew-tree) contribute to the immobilization of OM in the B horizon during podzolisation. On the other hand, the preferential path of water flow opened by the decayed taproot speeded up the degradation of the top of the B

horizon. As a result, taproots are involved in both the formation and degradation of the podzol Bh-horizon. We consider that root-mediated hydrologic processes that influence both, the formation and degradation of the Bh horizon, lead to “divergent pedogenesis”, a situation in which the spatial variability of the soil increases over time due to individual trees. (Phillips

2017). Because the cashew tree in the PC1 profile (Fig. 5b) is younger than 1950 (^{14}C ages = 118.2 ± 0.3 pMC), we posit that the morphological transformations caused by this tree-root occur fast in tropical podzols. Since the cashew trees are rather short-lived (~ 50 years), there will be likely a cumulative effect of many trees on well-developed podzols. The process of soil profile morphology alteration due to tree roots is local, thus, we must assume a short re-occurrence time interval during the last 6000 years (age of the island) that explain a wider scale changes of the entire cliff.

Conclusions

Changes in soil drainage conditions triggered by bioturbation (e.g., tree-roots) have a tremendous impact on the morphology of the studied podzols in a short temporal scale (~ 50 years). Our study indicates that the growth of taproots alters local soil porosity and permeability creating preferential water paths that remove or add OM in the B-horizon of tropical podzols. Such removal and addition of OM in the subsoil cause abrupt modifications in the shape and boundaries of the E and B horizons of podzols at an unexpectedly short timescale of a few decades. These modifications caused by individual taproots resulted in adjacent podzol profiles that significantly differ from each other in terms of the E and B horizons thicknesses, depths, and boundaries. Our inference has implications for land survey and biogeochemical studies. To ascertain the presence of podzol B-horizons during a soil survey in tropical rainforest, one must recognize that individual trees have the potential to overwhelming alter the depths and shapes of the B horizon within a short lateral distance. Furthermore, the mechanical effects of individual tree-roots on soil permeability indicate that the magnitude of inputs and outputs of DOM and metals in the B horizon can be highly modified within short temporal and spatial scales in tropical podzols.

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